Morpho Software Inc.
3944 State Street, Suite #340
Santa Barbara, CA 93105, USA
Tel: 805-569-9022
Fax: 805-569-5382
URL: http://www.morphosoft.com

MORF: Objectionable Image and Website Filter

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The Market

In the past, objectionable image filters – "porn" filters – were mainly used at home by parents to keep children away from pornographic materials on the Internet. While this is still a major use of filters – and indeed this demand is greater than ever – there is now a growing need for this product in corporations, schools and other public locations.

As Internet usage in the workplace has increased, companies have started to have to deal with issues such as loss of productivity, unnecessary bandwidth usage and legal liability. According to recent studies, for each worker that uses the Internet for private use during office hours, a company can lose between $30 to $100 ***per employee, per hour***. Given that 70% of visits to pornographic websites occur between 9am and 5pm on weekdays, this can add up to a great deal of money, especially for large firms.[1] Corporations can improve productivity and reduce non-business bandwidth usage by installing filtering software. Furthermore, companies are finding themselves legally liable for material that users download onto their computers. To protect themselves from harassment lawsuits, and to refute claims that they allow access to legally defined "addictive" materials (i.e., pornography), corporations can use filtering software and thus further decrease their non-business costs. Because of these issues, the corporate market for objectionable image filters is large and is getting larger, as indicated by a recent study from the IDC, which reported that Worldwide Employee Internet Management revenue reached $124 million in 2000, representing a 101% growth over 1999, and is forecasted to grow to $655 million by 2005, representing a 39% CAGR.[2]

In addition, as mentioned above, there has always been a strong demand for screening out objectionable information on the Web from underage children, and this demand is also growing. In December 2000, the *Children's Internet Protection Act (CIPA)* was signed into law in the United States, and went into effect in April 2001. CIPA requires the nation's 16,000 public libraries and 84,000 public schools that receive federal funding to equip their computers with filtering software to block access to visually pornographic Web sites, or risk losing their grant money. Also, most ISPs are looking to incorporate filtering technology into their offerings, as a way to help parents protect their children.

Given these trends, it is evident that the market for effective, high-quality filtering technology is substantial.

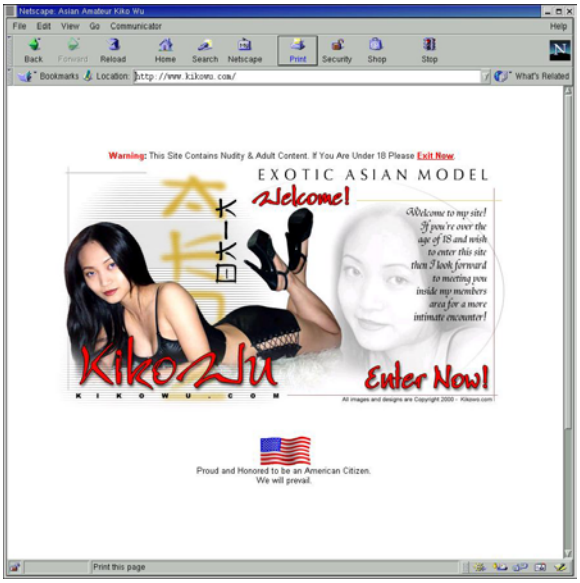
Product Description Using our patent-pending technology to analyze images on more than 140 parameters, our **M**ulti-modal **O**bjectionable info**R**mation **F**ilter (MORF) can carry out effective real-time filtering of pornographic content and websites. Because of the extent of the analysis it performs, MORF is able to differentiate between a nude picture on a pornographic site and a nude picture in a work of art, among other things. This particular feature is one that is inadequate or completely lacking in other filters. The product is also designed to minimize the amount of objectionable images that get through to the user, while also minimizing the amount of acceptable material that does not get through. In addition, MORF offers tremendous flexibility and can be customized and personalized to meet different filtering criteria. Because of these factors, and because it is simple to install and use MORF, the product is ideal for corporations, ISPs, portals and search engines in order to help them enforce Internet usage policies, manage bandwidth and ensure an adequate level of protection without undue cost, inconvenience or interference. This software is also computationally efficient to allow client-side, real-time filtering for home users.

Product Demonstration

The following three screens provide an illustration of how MORF works when browsing the Internet (regardless of which browser is used). When one attempts to access the pornographic site illustrated in Screen 1 below, MORF performs the following operations:

1. When downloading the home page, MORF processes all images and concludes that there is no pornographic image. Therefore, no image is blocked (as indicated in Screen 1).
2. After a user clicks on the "Enter Now" button, MORF identifies two pornographic images on the next page and therefore blocks them both from display. MORF replaces the identified objectionable images with patterned tiles (as shown in Screen 2). Note that the face and the body of the naked girl on the left-hand side of the page are two different images, and MORF correctly blocks the naked body but allows the face to be shown.
3. At this point, MORF determines that this site is indeed pornographic. If the user tries to click on any button in the menu bar, or tries to visit any other page on this site, a screen indicating "access denied" will be displayed (as shown in Screen 3). Furthermore, the URL of this site is included in the "blacklist" and further access to this site – even the home page – is completely denied.

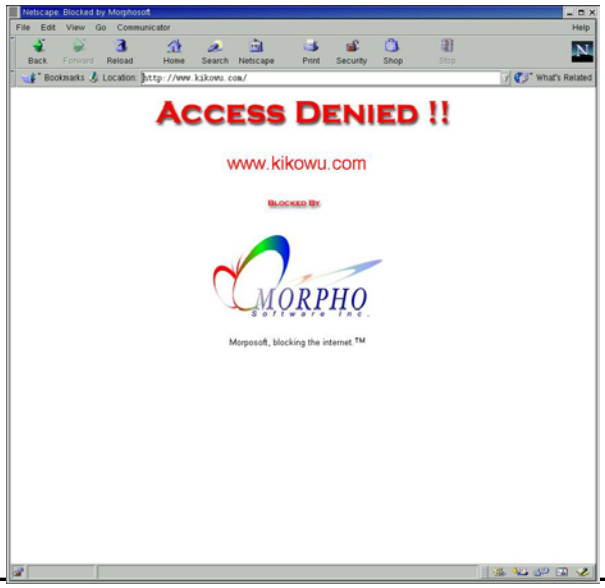
Screen 1



Screen 2



Screen 3



Competitive Product Comparison

The following table summarizes the key players in the objectionable image/website filtering market, and classifies the type of technology each uses. We are confident, however, that MORF is superior to any existing filtering software, because of our patent-pending technology (as discussed above, in Product Description). As shown below, other solutions either: 1) analyze the text naming and surrounding the images, 2) maintain a list of blacklisted Web addresses against which they match requested URLs or 3) analyze skin-color tone. Such solutions have encountered various known limitations which result in very poor filtering quality. For example, analyzing text often applies to English sites only. Furthermore, much of pornography's lexicon also has non-pornographic uses (e.g., wren tits, wood screws, breast cancer). Moreover, porn sites seem to incorporate more and more text directly into image files, which makes text analysis increasingly difficult and less accurate. On the other hand, maintaining a list of blacklisted Web addresses requires significant human resources and is not scalable due to the ongoing new 300-400 pornographic sites springing up daily on the Internet.[3] Common knowledge is that such a list from any current vendor is always out-of-date and only approximately 30 to 50 percent accurate. Finally, using skin tone analysis not only brings up the issue of what, exactly, is the color of skin, but also makes it difficult to differentiate between pornographic nudity and nudity in art or medical information.

COMPETITION: OBJECTIONABLE IMAGE/WEBSITE FILTERING	
COMPANIES/SOFTWARE	FILTERING TECHNOLOGY USED
CyberPatrol, CyberSitter, NetNanny, Cyber Sentinel, SurfWatch, N2H2, Stop Watch, Websense, etc.	Text-based (uses keywords) + URL list-based
eVision/eVe	Content-based (analyzes skin tone)
Morpho/MORF	Perception-based w/personalization

Company Background

Morpho Software is a start-up company aimed at providing customers with a variety of fast, effective, easy-to-use, and seamlessly integrated multimedia filtering, retrieval and organization solutions that satisfy their specific information management needs.

Founded by scientists from Univ. of California, Santa Barbara and Stanford who have extensive experience in analysis, indexing and retrieval of multimedia information, Morpho has innovative and patent-pending technologies that support a wide range of applications in multimedia information filtering and retrieval.

Morpho's products are based on proprietary breakthrough technologies in:

- Intelligent learning algorithms to understand image contents, and
- Innovative image characterization for effective image data representation

These technologies have resulted in one pending patent and four provisional patents.

[1] "Sex Tracker" study, Cyberworld, August 2001

[2] *Employee Internet Management 2001: Getting More Out of Less*, IDC, August 2001.

[3] *Tools and Strategies for Protecting Kids from Pornography and Their Applicability to Other Inappropriate Internet Content,* National Research Council White Paper.